SWINBURNE UNIVERSITY OF TECHNOLOGY

Alliance with FPT Education

VISUALIZATION CRITIQUE ASSIGNMENT

FROM HOSPITAL COUNTS TO RESPONSIBLE DECISIONS: A CRITIQUE OF THE HOSPITALISED INJURIES DASHBOARD



COS30045
DATA VISUALIZATION

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I. Introduction

The Australian Government's *Hospitalised injuries from road crashes* dashboard presents hospitalised road traffic injuries in Australia between 2011 and 2021. It combines case counts, rates per 100,000 population and average bed days with comparisons by age, sex, road-user category and residential remoteness (Department of Infrastructure, Transport, Regional Development, Communications and the Arts [DITRDCA], n.d.). Together, these indicators help readers examine the scale and distribution of hospitalised injuries, although each answers a different question.

The dashboard's functions suggest that road safety analysts and policy officers are a principal audience, alongside health planners and the public. Analysts may compare groups or trends, while health planners consider case volumes and hospital care. These uses align with national efforts to improve non-fatal crash measurement (Austroads, 2023). Specialists need comparable measures and explicit definitions; less experienced readers also need to distinguish large counts from high population rates. Users may ask which age and road-user groups account for the most hospitalisations, whether residential remoteness rankings change when population rates replace counts, and how hospitalisations have changed over time. They also need to know whether apparent differences reflect injury patterns, population size or changes in hospital reporting. These tasks guide this evaluation, following Munzner's (2009) emphasis on matching visualisation to users' needs.

This critique argues that the dashboard supports useful exploration but places too much responsibility on readers to connect its measures with their methodological limits. The analysis first establishes what the data represent, then examines whether the visual design supports valid comparisons and accessible interpretation. The recommendations address these connected issues by bringing complementary measures, essential qualifications and accessible reading routes closer to the evidence they explain.

II. Data Processing and Governance

A. Defining Cases and Hospital Care

The source Excel workbook published by BITRE in September 2023 contains aggregated hospital data for 2011–2021 (Bureau of Infrastructure and Transport Research Economics

[BITRE], 2023). Hospital records originate from the Australian Institute of Health and Welfare's National Hospital Morbidity Database; population data are attributed to the Australian Bureau of Statistics. The release date must therefore be distinguished from the years measured. Appendix A summarises the source and classifications relevant to this critique.

The Data Dictionary defines the relevant traffic injuries and exclusions used to reduce repeated case counting across episodes of care (DITRDCA, 2023). Selecting the correct injury category is therefore essential: summing every cause in the workbook would overstate the dashboard's scope. Cases and bed days also measure different aspects of hospital burden. A transfer may contribute additional days of care without adding another injury case, so discarding rows without a case count could remove relevant bed days. Similarly, average bed days describes mean care duration rather than the experience of every patient. The 2021 total of 198,934 bed days divided by 39,505 hospitalisations gives approximately 5.04, consistent with the displayed rounded value of 5 (Appendix B). This agreement supports the indicator's plausibility without establishing its internal calculation. It also illustrates why a rounded summary should not be interpreted as a typical stay of exactly five days.

B. Coverage, Exclusions and Accountability

The 2021 remoteness comparison demonstrates how aggregation affects interpretation. Its three displayed groups total 38,876 cases, whereas the overall indicator reports 39,505. The workbook identifies the remaining 629 as Other or unknown (Figure 1). The difference is therefore explained by category exclusion. However, readers must connect the chart's qualification with the overall indicator to reconcile their totals. Showing both together without an equally visible explanation can make consistent data appear contradictory.

The classifications also restrict the conclusions that can be drawn. Remoteness refers to usual residence, so the chart cannot identify dangerous crash locations. Hospitalisation data omit injuries treated entirely outside hospital. The unknown sex category includes intersex or indeterminate records, meaning that male and female do not exhaust the classification (DITRDCA, n.d.). Uneven missingness could distort group representation, but its presence has not been established here. These limits warrant explicit qualifications rather than

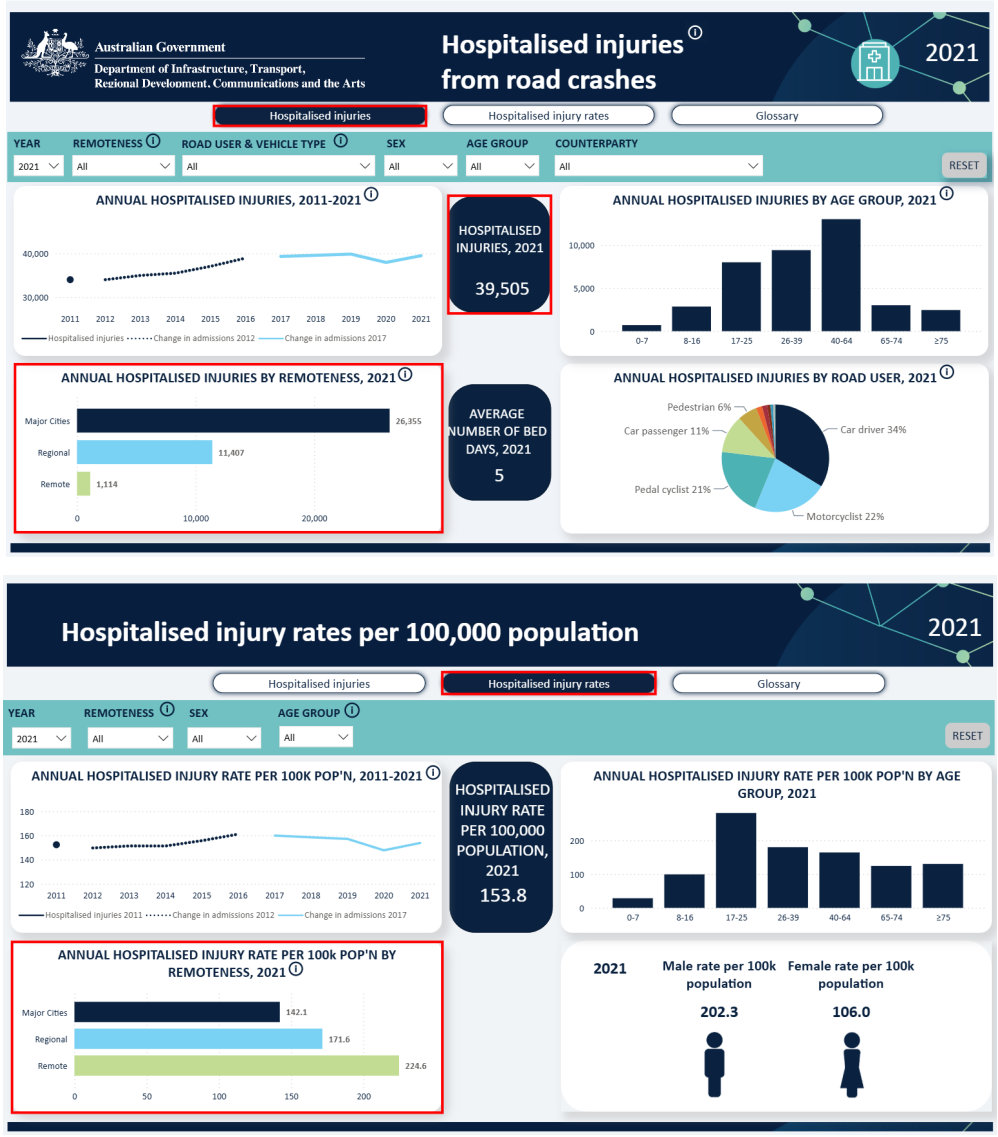


Figure 1. Residential remoteness counts and rates, showing the difference between displayed groups and the overall total. YEAR=2021, other filters All. Sources: DITRDCA (n.d.) and BITRE (2023).

assumptions about unobserved groups.

This has an ethical consequence: a smaller or excluded category should not be interpreted as having no need for services. Visibility in the report depends on collection and classification decisions as well as the underlying injury burden. The Glossary acknowledges exclusions, but their separation from the relevant comparisons weakens their practical role in preventing such interpretations.

Providing an aggregated workbook and Data Dictionary supports scrutiny without publishing individual patient records. Nevertheless, aggregation alone does not establish disclosure protection for every filtered combination. Reproducibility is also incomplete: the workbook

lacks the population-denominator tables needed to independently recalculate the displayed rates. Accessible documentation is therefore a strength, but the available release does not fully expose how every indicator is produced.

III. Visualisation Design

A. Comparing Case Volume and Population Rates

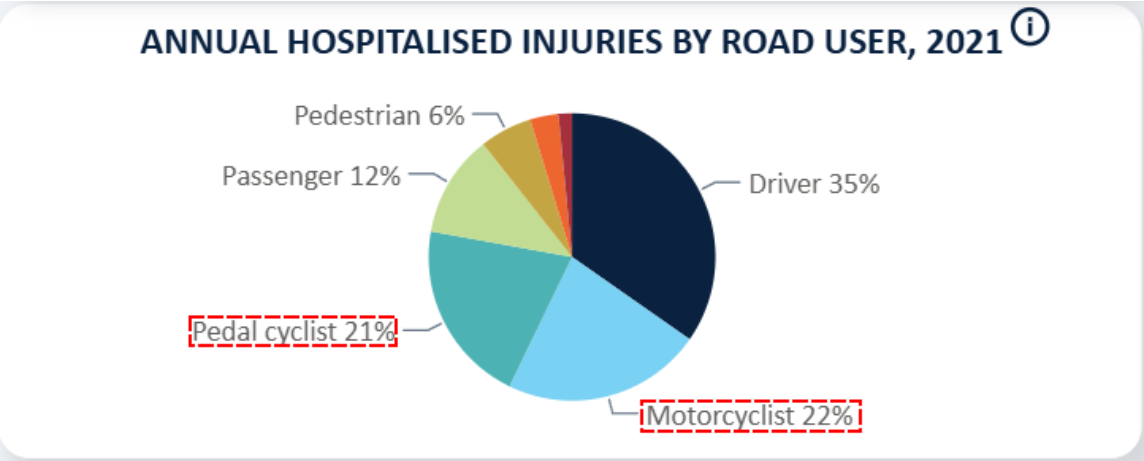
The distinction between counts and rates is the dashboard's strongest analytical feature. In 2021, Major Cities records 26,355 cases compared with 1,114 in Remote, yet their displayed rates are 142.1 and 224.6 per 100,000 respectively (Figure 1). Counts compare case volumes among resident groups; rates express frequency relative to population size. These perspectives may inform different priorities, and neither measures risk per trip or kilometre travelled. Similarly, ages 40–64 lead case counts, whereas 17–25 lead displayed rates. Unequal age-band widths further complicate interpreting count rankings as differences in individual risk.

The bar charts support magnitude comparison through length from a common zero baseline. Chronological age order preserves demographic context. However, separating counts and rates across pages requires readers to remember values and rankings while navigating, making the relationship between these measures harder to assess.

The road-user pie presents another comparison problem. Its labels identify Motorcyclist at 22% and Pedal cyclist at 21%, but the similar slices offer little visual separation (Figure 2). The pie communicates composition, while aligned bars would better support judging close magnitudes through a shared baseline, consistent with Cleveland and McGill (1984). The existing accessible data table supplies exact counts and deserves recognition; it complements the chart without resolving the difficulty of visual comparison.

B. Reading Trends Across Reporting Changes

The annual line chart maps time to horizontal position and magnitude to vertical position, making the sequence of changes easy to follow. Different colours and line styles distinguish reporting periods, signalling that the series requires care (Figure 3). This is a useful safeguard. However, the explanation of admission-rule changes in Victoria from July 2012 and New South Wales from June 2017 sits in the Glossary (DITRDCA, n.d.). Readers can encounter the apparent trend before understanding why some periods are not directly comparable.



Vehicle and road user	Hospitalisations	Year
Car driver	13,327	2021
Motorcyclist	8,883	2021
Pedal cyclist	8,163	2021
Car passenger	4,495	2021
Pedestrian	2,334	2021
Car unknown seat	717	2021
Other or unknown	601	2021
Heavy transport driver	383	2021
Pick-up truck or van occupant	341	2021
Bus occupant	169	2021
Heavy transport passenger	56	2021
Heavy transport unknown position	36	2021

Figure 2. Road-user composition and the detailed vehicle/road-user table. The table uses finer categories than the grouped pie. YEAR=2021, other filters All. Source: DITRDCA (n.d.).

A vertical scale above zero is defensible for a line chart intended to reveal annual variation. The greater interpretive concern is whether changes in recording practice can be mistaken for changes in road safety. Period styling acknowledges the problem but does not explain its cause at the point of comparison. The chart therefore provides historical context more effectively than methodological context.

Interaction requires clarity. Changing YEAR updates selected-year summaries while retaining the full 2011–2021 annual series. This preserves a useful overview, but readers must distinguish the selected year from the historical range. RESET provides a recovery route; explicit labels would clarify which views respond.

Appendix C records the tested interactions and remaining limits. Loading efficiency, complete

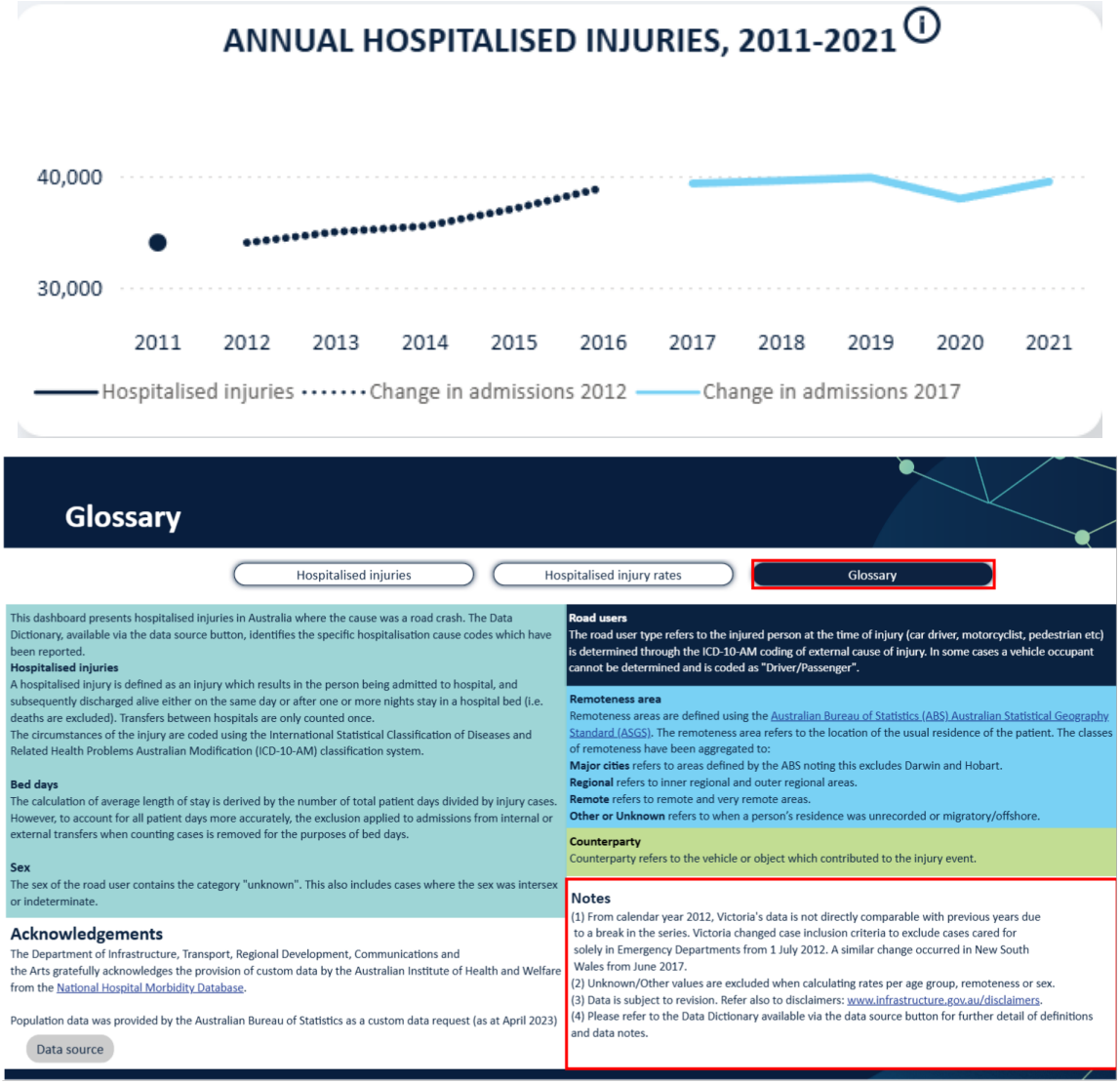


Figure 3. Annual hospitalised injuries, 2011–2021, and Glossary notes explaining changes to admission rules. Source: DITRDCA (n.d.).

drill-through behaviour and screen-reader operation have not been established.

C. Accessing the Evidence and Its Explanation

These interpretive demands become harder when essential information is difficult to access. At a viewport width of 320 CSS pixels, the Glossary scales to 25% rather than reflowing its text (Figure 4). The resulting small text obstructs the definitions on which responsible interpretation depends. This observation concerns the tested viewport, rather than every mobile device.

The annual cyan line also has a measured contrast ratio of 1.72:1 against white, below WCAG 2.2's 3:1 requirement for essential graphical objects (World Wide Web Consortium [W3C],

2023, Success Criterion 1.4.11). Its visibility can therefore be limited for readers with low vision. Keyboard operation and the accessible table provide valuable alternatives, although generic object names and an indirect focus sequence make orientation harder. As Chartability emphasises, accessibility involves navigating and understanding a visualisation as well as perceiving its marks (Elavsky et al., 2022). Here, access to methodological context is part of access to the analysis itself.

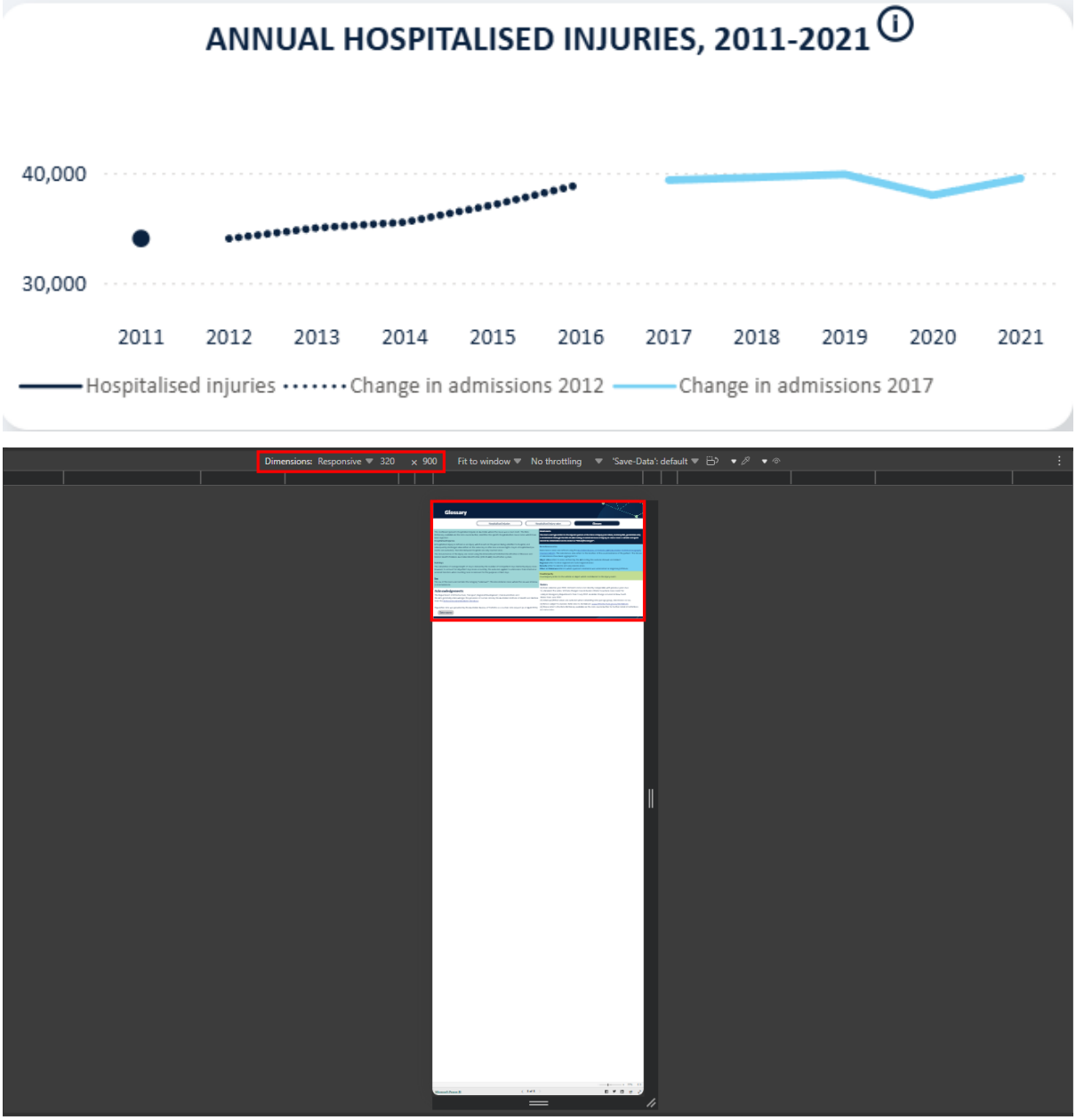


Figure 4. Annual line and Glossary at a 320-by-900 CSS pixel viewport. The contrast measurement applies to the tested line colour. Source: DITRDCA (n.d.).

IV. Recommendations

A. Bring Complementary Measures Together

The first priority is to place count and rate comparisons for age and residential remoteness in adjacent panels. Each pair should share category order and filter context, with separate, clearly labelled axes and units. Readers could then identify the largest case volume and highest population rate within one view. Chronological order should remain for age groups; the purpose is to clarify the meaning of each comparison, rather than suggest that one ranking determines policy priorities.

For road users, horizontal bars sorted by value and labelled with counts and percentages would make close differences easier to compare. The existing data table should remain available for exact lookup. Selected-year summaries should also identify their year clearly, while the annual chart states its retained historical range. These changes address comparison and navigation together by reducing the need to remember values or infer filter behaviour.

The panels should retain independent scales because counts and rates have different units. Matching their physical bar lengths across panels would falsely imply numerical equivalence. Consistent category placement, rather than a shared quantitative axis, should provide the link between them.

B. Explain Scope Where Conclusions Are Formed

The second priority is to attach essential qualifications to the affected evidence. The remoteness chart should identify usual residence and display included and excluded totals. At the default selection, “38,876 cases shown; 629 Other/Unknown cases excluded” would explain the relationship to the overall indicator. Both values must update with filters.

Short annotations at the relevant reporting boundaries should explain the admission-rule changes, while retaining existing period distinctions and linking to the full Glossary. An adjusted continuous trend should not be introduced without a defensible method. Concise notes should likewise distinguish population rates from travel-exposure risk and cases from days of care. Publishing the corresponding denominator tables would additionally allow independent rate checks. Together, these changes make limitations visible during interpretation and improve accountability for the reported measures.

C. Make the Main Reading Route Accessible

The third priority is to improve access within the report's normal reading sequence. Essential lines need sufficient contrast while retaining distinguishable styles. Meaningful control names, a logical focus order and a clearly signposted data-table route would improve orientation. These are feasible report-authoring measures supported by Microsoft's (2026) accessibility guidance.

Glossary text should remain readable on narrow screens. If the fixed report canvas prevents reflow, a responsive companion page could provide the full definitions, while short qualifications remain beside the charts. Text descriptions should communicate important comparisons and their limits rather than repeat titles, drawing on Lundgard and Satyanarayan (2022).

These changes preserve useful measures while strengthening their interpretation. Their success should be assessed through whether readers can distinguish the highest count from the highest rate, explain excluded cases and recognise reporting breaks through accessible reading routes. These are intended outcomes, not demonstrated usability improvements.

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Appendix A. Data Definitions

Source: BITRE's September 2023 workbook and Data Dictionary, covering 2011–2021 (BITRE, 2023; DITRDCA, 2023). Dashboard definitions are from its Glossary (DITRDCA, n.d.).

Term	Meaning in this report
Source rows	Aggregated combinations of year, month and demographic/injury categories, rather than individual patients. Numerical checks use the Traffic category.
Cases and bed days	These measure different things. Some rows without case counts still contribute bed days; removing those rows would discard care data.
Remoteness	Usual residence, not crash location. Regional combines Inner/Outer Regional; Remote combines Remote/Very Remote. Other/Unknown is excluded from the chart.
Unknown sex	Includes intersex or indeterminate records; male and female do not cover the entire classification.
Reporting breaks	Admission rules changed in Victoria from July 2012 and New South Wales from June 2017. Comparisons across these dates need qualification.
Population rates	ABS population data are acknowledged, but denominator tables are absent from the workbook. Rates quoted here were read from the dashboard, not independently recalculated.

Appendix B. Checks on the 2021 Figures

These checks use YEAR=2021, other filters All, and the Traffic category. Counts were checked against the published workbook (BITRE, 2023); displayed rates and pie labels come from the dashboard (DITRDCA, n.d.).

B1. Remoteness totals

Group	Cases	Rate per 100,000
Major Cities	26,355	142.1
Regional	11,407	171.6
Remote	1,114	224.6
Other or unknown	629	Not shown
Overall	39,505	153.8

The three displayed groups total 38,876. Adding the 629 Other/Unknown cases gives 39,505, matching the overall indicator. Subgroup rates cannot be added because their population denominators differ.

B2. Average bed days

The workbook totals give **198,934 / 39,505 = 5.035666… days**, consistent with the displayed rounded value of 5. This agreement does not confirm the dashboard's internal formula or describe the spread of hospital stays.

B3. Grouped pie and detailed table

Figure 2 shows a grouped pie alongside a table with finer categories. The following sums explain the different labels.

Pie group	Categories in the detailed table
Driver: 13,710	Car driver 13,327 + Heavy transport driver 383.
Passenger: 4,551	Car passenger 4,495 + Heavy transport passenger 56.
Driver/Passenger: 1,263	Car unknown seat 717 + Pick-up truck or van occupant 341 + Bus occupant 169 + Heavy transport unknown position 36.

Motorcyclist (8,883) and Pedal cyclist (8,163) retain the same categories. Their shares of 39,505 round to the pie's 22% and 21% labels.

Appendix C. Interaction and Accessibility Checks

Recorded on 15 October 2026 using the assigned dashboard (DITRDCA, n.d.). The screenshots supplied on 18 October illustrate the report; they do not replace the earlier measurements.

Check	Observed result
YEAR	On the rates page, 2021 to 2020 changed the overall rate from 153.8 to 147.7. Selected-year labels changed; the annual chart retained 2011–2021.
RESET	Restored YEAR=2021 and the overall rate of 153.8 in the tested sequence.
Data table	Alt+Shift+F11 opened the road-user pie's HTML table with headers and counts.
Keyboard orientation	Names included `Image` and `3 items`; focus moved among notes, summaries and grouped objects.
Line contrast	Authored colours #79D1F4 and #FFFFFF give 1.72:1, below the 3:1 criterion for essential graphics (W3C, 2023).
Narrow viewport	At 320×900 CSS pixels, the Glossary canvas scaled to 25% without text reflow.

These were technical checks, not participant testing. Screen-reader output, complete filter/drill-through behaviour and loading performance remain unverified. The results do not establish WCAG compliance for the whole dashboard.